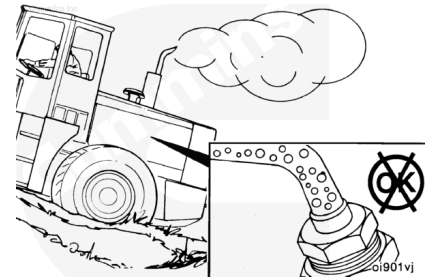


Trainings ▾

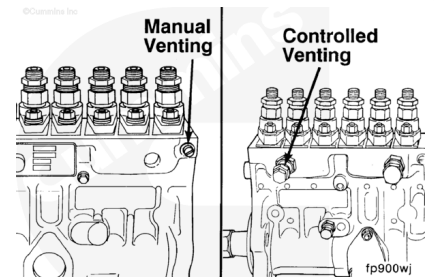
General Information

Air from uncorrected leaks in the supply circuit will make the engine:

1. Difficult to start
2. Run rough
3. Misfire
4. Produce low power
5. Emit excessive smoke
6. Produce fuel knock.

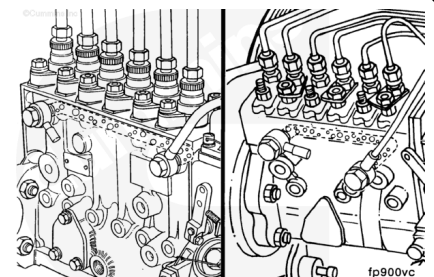


The Nippondenso™ EP-9 in-line fuel injection pumps will require additional venting prior to initial start-up, pump replacement, or if the engine has been run out of fuel.



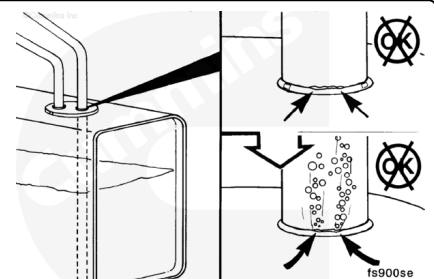
The MW, A, and P fuel injection pumps equipped with engine-side fuel drain arrangement create a self-bleeding system for air introduced during replacement of the supply-side components.

Note : For faster air purge, small amounts of air can be bled from the pump by operating the hand primer on the fuel transfer pump or by cranking the engine.



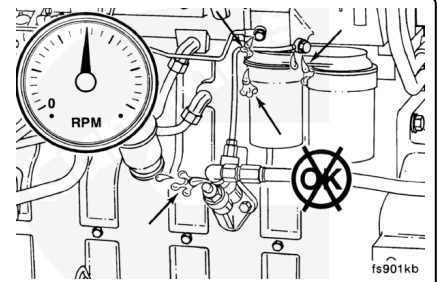
A source often overlooked for air to enter the fuel system is between the inlet of the prefilter and the suction tube in the fuel tank. Fuel tanks that have the outlet fitting at the top will have a suction tube that extends to the bottom of the fuel tank. Cracks or pin holes in the weld that join the suction tube to the fitting can let air enter the fuel system.

Check all the fittings from the fuel supply on the fuel tank to the fuel inlet of the fuel transfer pump; make sure they are all tight. Use a sight glass at the fuel transfer pump inlet to check



for air in the supply line.

Since the fuel transfer pump provides a positive pressure through the fuel filter and supply line to the fuel injection pump, loose connections or malfunctioning seals will show as a fuel leak and **not** as an air leak.

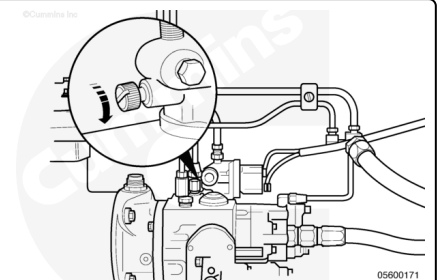


Vent

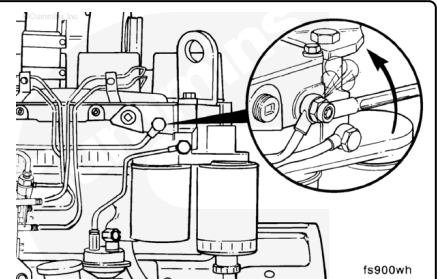
For some engine equipped with fuel shutoff valve:

Before venting the air in the fuel, turn the knob on the shutoff valve **clockwise** to open the valve.

After finish venting, turn the knob on the shutoff valve **counterclockwise** to close the valve.



Open the banjo bleed screw to bleed the low-pressure fuel line.

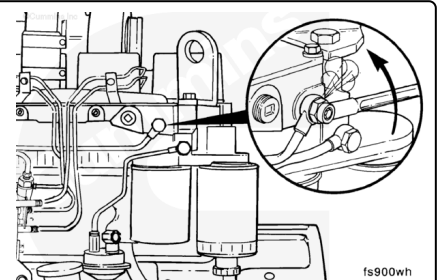


⚠ WARNING ⚠

The pressure of the fuel in the line is sufficient to penetrate the skin and cause serious injury.

To check for air in the high-pressure lines, loosen the fittings at the head. Crank the engine to allow entrapped air to bleed from the fuel line. Tighten the fittings.

Torque Value: 38 n•m [28 ft-lb]



⚠ CAUTION ⚠

Do not bleed a hot engine, as this could cause fuel to spill onto a hot exhaust manifold, creating a danger of fire.

Operate the engine and bleed one fuel line at a time until the engine runs smoothly.

